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DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

MICRO PROJECT

DESIGN, SIMULATION, AND HARDWARE IMPLEMENTATION OF AN NE555-BASED PWM DC MOTOR SPEED CONTROLLER

PBEET304 ANALOG ELECTRONICS

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ABSTRACT

Precise and energy-efficient speed regulation of Direct Current (DC) motors is a foundational requirement in modern industrial automation, robotics, electric traction, and consumer appliances. Conventional resistive or linear voltage-dropping speed control methods suffer from severe power dissipation in the form of heat and substantial loss of starting torque at low operating speeds. Pulse Width Modulation (PWM) overcomes these limitations by applying the full supply voltage to the motor in rapid, discrete switching pulses, thereby controlling the average armature voltage while maintaining high efficiency and strong low-speed torque. The objective of this micro project is to design, mathematically analyze, simulate in LTspice, and implement a practical constant-frequency PWM DC motor speed controller using an NE555 monolithic timer IC and an N-channel enhancement-mode power MOSFET.

In the proposed system, the NE555 timer (U_1) is configured as a diode-steered astable multivibrator powered by a 12 VDC supply (V_1), utilizing two steering diodes (D_1, D_2) and a 47 kΩpotentiometer (U_2) to separate the charging and discharging paths of a 100 nFtiming capacitor (C_1). This topology ensures that the total resistance in the timing loop remains constant regardless of the potentiometer wiper position, yielding a stable switching frequency of approximately 300 Hzwhile enabling continuous duty cycle modulation from 2.1%to 97.9%. Comprehensive circuit protections and stability improvements were incorporated into the final design, including a 100 nFbypass capacitor (C_3) on the control voltage pin (Pin 5) to suppress electromagnetic interference (EMI), a 100 ΩMOSFET gate resistor (R_1) to limit capacitive inrush current to a safe peak of 120 mAand dampen parasitic LCringing, a 10 kΩgate-to-source pull-down resistor (R_3) to prevent floating-gate false triggering, a freewheeling diode (D_3) across the motor terminals to clamp inductive back-EMF spikes, and a 470 Ωindicator branch (R_4). LTspice transient simulations executed over a 0 msto 100 mswindow confirmed stable rectangular PWM output pulses V(n002)switching cleanly between 0 Vand 12 V, while at a 75%potentiometer wiper setting (wiper=.75), the steering diodes exhibited synchronous peak charging and discharging currents of $I(D_1) \approx 204$ μA(during the 12 VON state) and $I(D_2) \approx 600$ μA(during the 0 VOFF state), respectively.

The validated circuit was successfully assembled and tested on a physical breadboard powered by a 12 VDC adapter, demonstrating smooth, stepless motor speed control and synchronous LED brightness variation from near-cutoff to full rated speed. The close agreement between theoretical KVL/mesh calculations, LTspice simulation waveforms, and hardware measurements confirms the robustness, thermal efficiency, and practical viability of the designed analog PWM controller.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Direct Current (DC) motors are widely utilized across engineering applications ranging from precision robotic actuators and cooling fans to conveyor belts and automotive subsystems due to their straightforward torque-speed characteristics and ease of control [1], [2]. For a permanent-magnet or shunt-wound DC motor, the steady-state rotational speed $\omega$is directly proportional to the applied armature voltage $V_a$minus the armature resistive drop:

$$\omega \propto \frac{V_a - I_a R_a}{k\phi}$$

where $I_a$is the armature current, $R_a$is the armature winding resistance, and $k\phi$represents the back-EMF constant determined by the magnetic flux.

Historically, varying the applied voltage $V_a$was achieved by inserting a variable series resistor (rheostat) or operating a power transistor in its linear active region. However, both methods introduce severe engineering drawbacks:

1. High Power Loss: The series dropping element continuously dissipates power equal to $P_{loss} = (V_{CC} - V_a) \cdot I_a$, converting a large fraction of source energy into waste heat [3].
2. Loss of Low-Speed Torque: At low voltage settings, the large series resistance limits the starting and stall current required to overcome static friction, causing the motor to stall easily under mechanical load.

1.2 PULSE WIDTH MODULATION (PWM) PRINCIPLE

Pulse Width Modulation (PWM) resolves these inefficiencies by operating the power semiconductor switch strictly in two states: fully ON (ohmic saturation region) and fully OFF (cutoff region) [3], [4]. Instead of reducing the amplitude of the supply voltage, the controller rapidly switches the full supply voltage V_{CC}across the motor terminals at a fixed time period $T = T_{ON} + T_{OFF}$.

The ratio of the ON duration (T_{ON}) to the total switching period (T) is defined as the Duty Cycle (D):

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100\%$$

The effective average DC voltage V_{avg}delivered to the motor armature is:

$$V_{avg} = D \cdot V_{CC} = \left(\frac{T_{ON}}{T_{ON} + T_{OFF}}\right) V_{CC}$$

Because the mechanical rotor possesses physical inertia and the motor windings possess inductance, the motor acts as a natural low-pass filter when the switching frequency is sufficiently high (typically $> 100\ \mathrm{Hz}$). Consequently, the motor responds smoothly to the average voltage V_{avg}while receiving full-voltage pulses ($12\ \mathrm{V}$) that preserve strong starting torque even at low speeds. Furthermore, because the MOSFET switch exhibits nearly zero current when OFF ($I_D \approx 0\ \mathrm{A}$) and near-zero drain-to-source resistance when fully ON ($R_{DS(on)} \approx 17.5\ \mathrm{m\Omega}$), switching power dissipation is negligible.

1.3 OBJECTIVES OF THE WORK

The primary objectives of this micro project are:

1. To analyze and design a constant-frequency, variable-duty-cycle astable multivibrator using the NE555 timer IC (U_1) and a diode-steered potentiometer network (D_1, D_2, U_2).
2. To eliminate common schematic flaws by incorporating proper Pin 5 (`CV`) noise decoupling (C_3), MOSFET gate current-limiting (R_1) and pull-down (R_3) resistors, inductive flyback protection (D_3), and visual PWM feedback (R_4, LED).
3. To perform theoretical KVL/mesh timing and current analysis and verify the circuit behavior using LTspice transient simulations (`MY PROJECT (PWM).asc`), correlating the output voltage waveform V(n002)with the diode currents $I(D_1)$and $I(D_2)$.
4. To assemble, debug, and test the physical circuit on a breadboard powered by a 12 VDC adapter and evaluate its real-world performance.

CHAPTER 2: LITERATURE REVIEW

The evolution of analog pulse width modulation circuits and solid-state DC motor drives has been extensively documented in electronics literature over the past five decades.

In 1972, Hans Camenzind designed the monolithic 555 timer IC for Signetics Corporation, introducing an integrated architecture combining two voltage comparators, an internal $5\ \text{k}\Omega$resistive voltage-divider reference network, a set-reset (SR) flip-flop, and an open-collector discharge transistor [1]. As detailed in foundational analog electronics texts by Boylestad and Nashelsky [2], the standard astable configuration of the 555 timer charges an external timing capacitor Cthrough series resistors $R_A + R_B$and discharges it solely through R_B. A fundamental limitation of the standard astable circuit is that the charging time $T_{ON} = 0.693(R_A + R_B)C$is always strictly greater than the discharging time $T_{OFF} = 0.693R_BC$, restricting the achievable duty cycle to $> 50\%$and causing the oscillation frequency to shift substantially whenever $R_B$is adjusted.

To achieve full-range duty cycle control (0% to 100%) at a constant switching frequency, Sedra and Smith [3] and Rashid [4] describe the diode-steered astable topology. By placing two steering diodes in opposite orientations across the outer terminals of a potentiometer, the charging current is forced through one portion of the potentiometer while the discharging current is routed through the complementary portion. Since the sum of the two potentiometer segments remains constant across all wiper settings, the total period Tremains virtually invariant.

Further studies on MOSFET gate drive dynamics and inductive load switching highlight two critical practical considerations for 555-driven motor controllers [4], [5]:

1. Capacitive Gate Drive Transients: Power MOSFETs such as the IRFZ44N exhibit a significant input gate-to-source capacitance ($C_{iss} \approx 1470$ pF) that acts as a momentary short circuit during fast logic transitions. Without a series gate resistor, high peak currents can overload the NE555 output stage (rated at 200 mA) and excite high-frequency parasitic LCringing on the gate lead [3], [4].
2. Inductive Back-EMF Clamping: Switching off current in an inductive motor winding generates a high-voltage transient governed by $V_L = -L\frac{di}{dt}$. Incorporating a rectifier flyback diode across the inductive load provides a safe freewheeling path for the decaying armature current, preventing avalanche breakdown of the MOSFET [4], [5].

CHAPTER 3: PROPOSED SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE OVERVIEW

The complete proposed PWM DC Motor Speed Controller (matching the verified LTspice schematic `MY PROJECT (PWM).asc`) consists of three interconnected functional stages powered by a single $12\ \mathrm{V}$DC supply (V_1), where the negative terminal serves as the common $0\ \mathrm{V}$Ground reference:

1. Stage 1: Diode-Steered Constant-Frequency Timing Network ($R_2 = 1\ \mathrm{k\Omega}$, steering diodes $D_1$and D_2, potentiometer $U_2$with $R_{tot} = 47\ \mathrm{k\Omega}$, and timing capacitor $C_1 = 100\ \mathrm{nF}$).
2. Stage 2: NE555 Comparator and Oscillator Core (U_1 and Pin 5 control voltage bypass capacitor $C_3 = 100\ \mathrm{nF}/\ 10\ \mathrm{nF}$).
3. Stage 3: Power MOSFET Gate Driver, Flyback Protection, and Output Load Stage (gate series resistor $R_1 = 100\ \Omega$, gate pull-down resistor $R_3 = 10\ \mathrm{k\Omega}$, NMOS power transistor M_1, flyback diode D_3, indicator resistor $R_4 = 470\ \Omega$, bulk decoupling capacitor $C_2 = 10\ \mu\mathrm{F}$, and DC Motor U_3).

3.2 DETAILED CIRCUIT OPERATION

3.2.1 Why Trigger (Pin 2) and Threshold (Pin 6) Are Tied to the Timing Capacitor

Inside the NE555 IC (U_1), three matched $5\ \mathrm{k\Omega}$internal resistors form a voltage divider between V_{CC}(Pin 8) and GND (Pin 1), establishing two precise comparator reference voltages:

- Upper Comparator Reference (`THRS` - Pin 6): $\frac{2}{3}V_{CC}$(8.0 V for $V_{CC} = 12\ \mathrm{V}$).
- Lower Comparator Reference (`TRIG` - Pin 2): $\frac{1}{3}V_{CC}$(4.0 V for $V_{CC} = 12\ \mathrm{V}$).

By connecting `TRIG` (Pin 2) and `THRS` (Pin 6) directly together to the top plate of the $100\ \mathrm{nF}$timing capacitor (C_1) and the wiper of potentiometer U_2, both comparators continuously monitor the instantaneous capacitor voltage $V_{C1}(t)$, creating a self-sustaining oscillation loop:

1. Initial Power-Up & Charging State (T_{ON}): At $t = 0$, $V_{C1} = 0\ \mathrm{V} < \frac{1}{3}V_{CC}$. The Lower Comparator triggers the internal flip-flop, driving `OUT` (Pin 3, node `n002`) HIGH (12 V) and turning the internal discharge transistor at `DIS` (Pin 7) OFF (open circuit). Current flows from the $+12\ \mathrm{V}$rail through resistor $R_2(1\ \mathrm{k\Omega})$, forward-biases steering diode D_1, passes through the left segment of the $47\ \mathrm{k\Omega}$potentiometer (U_2), and charges capacitor $C_1(100\ \mathrm{nF})$.
2. Upper Threshold Transition: Once $V_{C1}(t)$rises to $\frac{2}{3}V_{CC}(8.0\ \mathrm{V})$, the Upper Comparator (`THRS`) resets the flip-flop. `OUT` (Pin 3, node `n002`) switches LOW (0 V), turning the MOSFET (M_1)

and motor (U_3) OFF, while `DIS` (Pin 7) switches ON, connecting Pin 7 internally to Ground (0 V).

3. Discharging State (T_{OFF}): Capacitor C_1now discharges from $\frac{2}{3}V_{CC}$(8.0 V) down to $\frac{1}{3}V_{CC}$(4.0 V) through the right segment of the 47 kΩpotentiometer (U_2), forward-biasing steering diode D_2, and sinking into `DIS` (Pin 7) to Ground.
4. Lower Trigger Transition: The instant $V_{C1}(t)$drops to $\frac{1}{3}V_{CC}$(4.0 V), `TRIG` (Pin 2) sets the flip-flop again, returning the circuit to the charging state.

Capacitor Voltage $V_{C1}(t)$Active Pin `DIS` (Pin 7) `OUT` V(n002)Capacitor Mode

Drops to $\frac{1}{3}V_{CC}$(4 V) `TRIG` (Pin 2) Open (OFF) HIGH (12 V) Charging (4 V → 8 V)

Rises to $\frac{2}{3}V_{CC}$(8 V) `THRS` (Pin 6) Grounded (ON) LOW (0 V) Discharging (8 V → 4 V)

3.3 MATHEMATICAL TIMING AND KVL MESH ANALYSIS

Let the total resistance of potentiometer U_2be $R_{tot} = 47\ \text{k}\Omega$, and let $w \in [0, 1]$represent the fractional wiper setting (`wiper = 0.75` in the LTspice schematic):

- Charging segment resistance (left side of U_2in series with D_1):
$$R_A = w \cdot R_{tot} = 0.75 \times 47\ \text{k}\Omega = 35.25\ \text{k}\Omega$$
- Discharging segment resistance (right side of U_2in series with D_2):
$$R_B = (1 - w) \cdot R_{tot} = 0.25 \times 47\ \text{k}\Omega = 11.75\ \text{k}\Omega$$
- Total potentiometer resistance for any wiper position w:
$$R_A + R_B = wR_{tot} + (1 - w)R_{tot} = R_{tot} = 47\ \text{k}\Omega$$

3.3.1 Loop 1: Charging Mesh Analysis (T_{ON})
During the charging interval ($V(\text{n002}) = 12$ V), Pin 7 (`DIS`) is open-circuited and diode $D_2$is reverse-biased. Applying Kirchhoff's Voltage Law (KVL) around the charging path from V_1(12 V) through R_2, D_1, R_A, and $C_1$to Ground:

$$V_{CC} - I(D_1) \cdot (R_2 + R_A) - V_{D1} - V_{C1}(t) = 0$$

where $R_2 = 1\ \text{k}\Omega$, $R_A = 35.25\ \text{k}\Omega$, and $V_{D1} \approx 0.6\ \text{V}$. Solving for the instantaneous charging diode current $I(D_1)$:

$$I(D_1) = \frac{V_{CC} - V_{D1} - V_{C1}(t)}{R_2 + R_A}$$

As $V_{C1}(t)$charges from $\frac{1}{3}V_{CC}(4.0\ \text{V})$ to $\frac{2}{3}V_{CC}(8.0\ \text{V})$ with time constant $\tau_{charge} = (R_2 + R_A)C_1$:

$$T_{ON} = 0.693 \cdot (R_2 + R_A)C_1 = 0.693 \times (1\ \text{k}\Omega + 35.25\ \text{k}\Omega) \times 100\ \text{nF} = 2.512\ \text{ms}$$

3.3.2 Loop 2: Discharging Mesh Analysis (T_{OFF})
During the discharging interval ($V(\text{n002}) = 0\ \text{V}$), Pin 7 (`DIS`) is internally shorted to Ground ($V_{DIS} \approx 0\ \text{V}$) and diode $D_1$is reverse-biased. Applying KVL around the discharging loop from C_1through R_B, D_2, and Pin 7 to Ground:

$$V_{C1}(t) - I(D_2) \cdot R_B - V_{D2} - V_{DIS} = 0$$

Solving for the instantaneous discharging diode current $I(D_2)$:

$$I(D_2) = \frac{V_{C1}(t) - V_{D2}}{R_B}$$

As $V_{C1}(t)$discharges from $\frac{2}{3}V_{CC}(8.0\ \text{V})$ down to $\frac{1}{3}V_{CC}(4.0\ \text{V})$ with time constant $\tau_{discharge} = R_B C_1$:

$$T_{OFF} = 0.693 \cdot R_B C_1 = 0.693 \times 11.75\ \text{k}\Omega \times 100\ \text{nF} = 0.814\ \text{ms}$$

3.3.3 Total Period, Constant Frequency, and Duty Cycle Equations
Summing $T_{ON}$and T_{OFF}gives the constant switching period T:

$$T = T_{ON} + T_{OFF} = 0.693 \cdot (R_2 + R_A + R_B)C_1 = 0.693 \cdot (R_2 + R_{tot})C_1$$

$$T = 2.512\ \text{ms} + 0.814\ \text{ms} = 3.326\ \text{ms}$$

The switching frequency $f$is therefore constant across all potentiometer settings:

$$f = \frac{1}{T} = \frac{1}{3.326 \times 10^{-3}\ \text{s}} \approx 300.6\ \text{Hz}$$

For `wiper = 0.75`, the theoretical duty cycle $D$is:

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100\% = \frac{2.512\ \text{ms}}{3.326\ \text{ms}} \times 100\% = 75.5\%$$

3.4 ENGINEERING PROTECTIONS AND COMPONENT SELECTION

1. Control Voltage (`CV` Pin 5) Bypass Capacitor (C_3):
   Connecting capacitor C_3between Pin 5 (`CV`) and Ground filters high-frequency switching

noise and motor brush EMI from corrupting the internal $\frac{2}{3}V_{CC}$ comparator reference, while Pin 1 (`GND`) connects directly via a solid wire to the 0 V ground rail.

2. MOSFET Gate Drive Network ($R_1 = 100\ \Omega$ and $R_3 = 10$ kΩ):
   The $100\ \Omega$ series gate resistor (R_1) limits the capacitive inrush current into the MOSFET gate to $I_{G(peak)} = \frac{12\text{ V}}{100\ \Omega} = 120$ mA (well below the 200 mA rating of the NE555) and dampens parasitic LC ringing. The 10 kΩ gate-to-source pull-down resistor (R_3) bleeds off stray electrostatic charge and keeps M_1 strictly OFF when unpowered.
3. Inductive Flyback Protection (D_3) and Visual Indicator (R_4 + LED):
   Diode D_3 (`1N4007`) is placed in reverse bias across the motor load (Cathode to $+12$ V, Anode to M_1 Drain) to safely circulate inductive back-EMF currents when M_1 turns OFF. Resistor R_4 ($470\ \Omega$) limits current through the parallel LED indicator to ≈ 21.3 mA at 12 V ($P_{R4} \approx 0.21\text{ W} < 0.25\text{ W}$).

3.5 BILL OF MATERIALS (MATCHING LTSPICE SCHEMATIC DESIGNATORS)

Designator Component Description Value / Specification

V_1 DC Power Supply $12\ \text{V}$ DC Adapter / Source

U_1 Monolithic Timer IC `NE555` (DIP-8)

U_2 Rotary Potentiometer $R_{tot} = 47\ \text{k}\Omega$ (`pot.sub`, `wiper=.75`)

U_3 DC Brushed Motor Load $12\ \text{V}$ DC Motor Subcircuit

M_1 N-Channel Power MOSFET `IRFZ44N` (`NMOS`, TO-220)

D_1, D_2 Steering Diodes `1N4148` / `1N4007`

D_3 Flyback Protection Diode `1N4007` ($1000\ \text{V}$, $1\ \text{A}$)

R_1 MOSFET Gate Series Resistor $100\ \Omega$, $0.25\ \text{W}$

R_2 Timing Pull-Up Resistor $1\ \text{k}\Omega$, $0.25\ \text{W}$

R_3 MOSFET Gate Pull-Down $10\ \text{k}\Omega$, $0.25\ \text{W}$

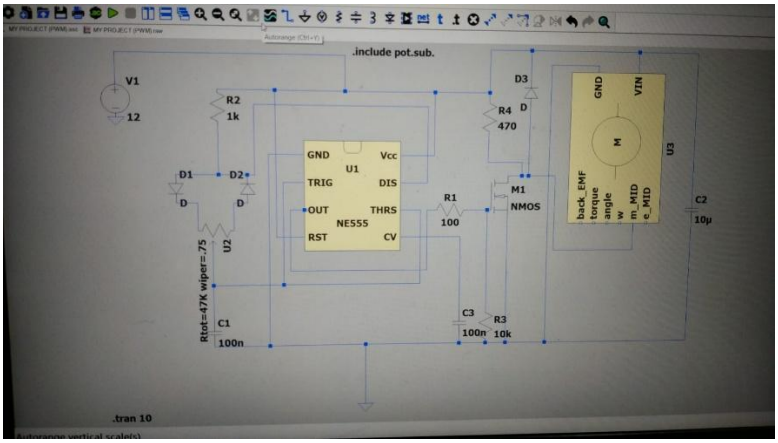
R_4 LED Indicator Series Resistor $470\ \Omega$, $0.25\ \text{W}$

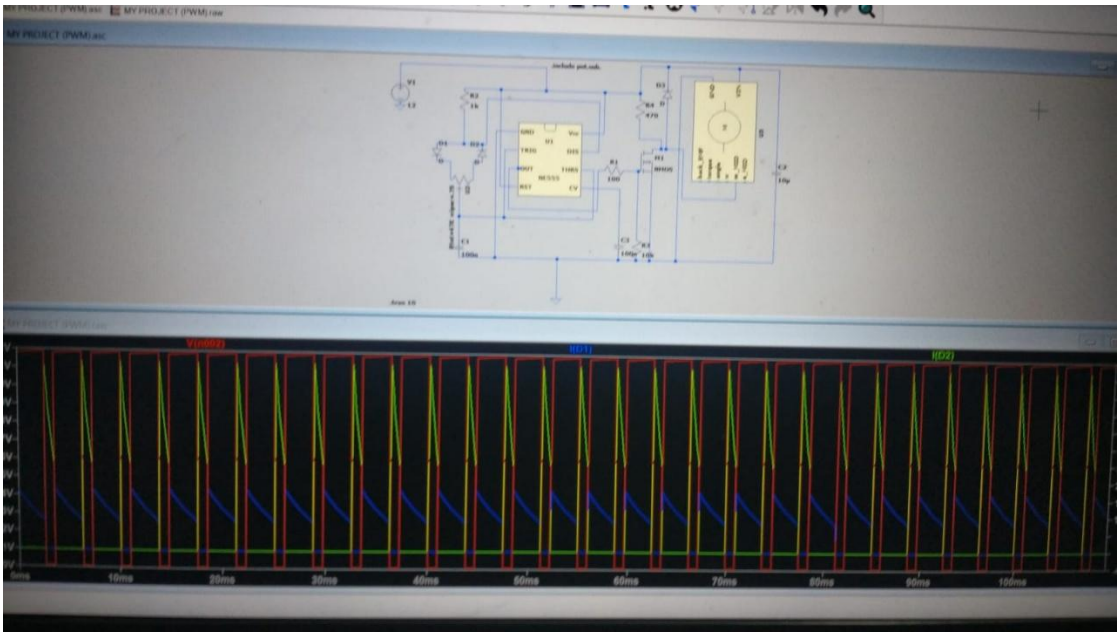
C_1 Timing Capacitor $100\ \text{nF}$ ($50\ \text{V}$ Ceramic, Code `104`)

C_2 Bulk Decoupling Capacitor $10\ \mu\text{F}$ – $100\ \mu\text{F}$ ($25\ \text{V}$ Electrolytic)

C_3 Pin 5 (`CV`) Bypass Capacitor $10\ \text{nF}$ – $100\ \text{nF}$ ($50\ \text{V}$ Ceramic)

CHAPTER 4: SYSTEM SIMULATION





4.1 LTSPICE SCHEMATIC TOPOLOGY (`MY PROJECT (PWM).asc`)

The circuit was simulated in LTspice XVII using the custom potentiometer subcircuit `.include pot.sub.` with `Rtot=47K wiper=.75`. In the motor subcircuit block U_3:

- `VIN` is connected to the $+12$ Vrail.
- `GND` is connected to the common 0 Vground rail.
- `m_MID` and `e_MID` are tied together and connected to the switched low-side node at the Drain of MOSFET M_1, the anode of flyback diode D_3, and the bottom of resistor R_4.
- `back_EMF`, `torque`, `angle`, and `w` serve as open-circuit diagnostic mechanical outputs.

4.2 THREE-TRACE TRANSIENT WAVEFORM ANALYSIS ($V(\mathrm{n002})$, $I(D_1)$, AND $I(D_2)$)

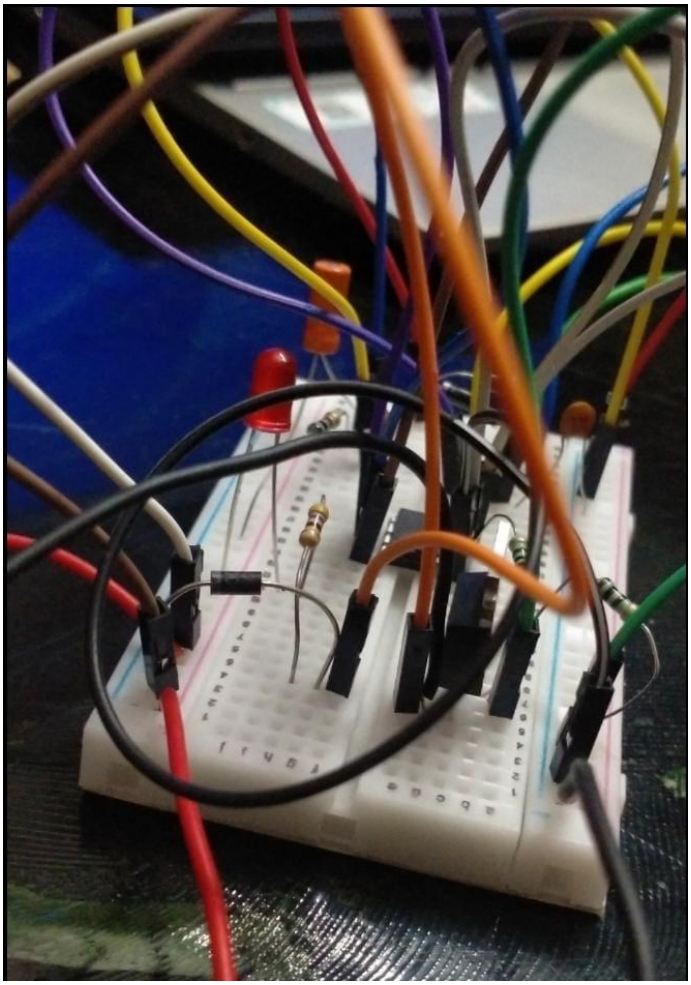
In the LTspice waveform viewer (`MY PROJECT (PWM).raw`), three simultaneous traces were plotted over a 0 msto 100 mswindow (displaying approximately 30complete PWM cycles):

1. Red Trace — Output PWM Voltage $V(\mathrm{n002})$:

- Represents the voltage at the `OUT` pin of the NE555 timer feeding gate resistor R_1.
- The waveform is a crisp, rectangular pulse train switching cleanly between 0 V(bottom axis) and 12 V(top axis) on the left vertical scale.
- For each 3.33 msperiod, V(n002)remains HIGH (12 V) for $T_{ON} \approx 2.51$ ms(75.5% of the cycle) and drops LOW (0 V) for $T_{OFF} \approx 0.81$ ms(24.5% of the cycle).

2. Blue Trace — Charging Diode Current $I(D_1)$:
   - Notice in the plot that the blue trace $I(D_1)$conducts strictly during the interval when the red trace V(n002)is HIGH (12 V).
   - At the rising edge of V(n002), when $V_{C1}$is at its lower threshold of 4.0 V, the charging current jumps to its theoretical peak:
   $$I(D_1)_{peak} = \frac{12.0\text{ V} - 0.6\text{ V} - 4.0\text{ V}}{1\text{ k}\Omega + 35.25\text{ k}\Omega} = \frac{7.4\text{ V}}{36.25\text{ k}\Omega} \approx 204.1\ \mu\text{A}$$
   - Across the 2.51 msHIGH pulse, $I(D_1)$decays exponentially down to $\approx 93.8\ \mu$Aas C_1charges to 8.0 V. The instant V(n002)switches LOW, D_1becomes reverse-biased and $I(D_1)$drops to $0\ \mu$A.
3. Green Trace — Discharging Diode Current $I(D_2)$:
   - Notice in the plot that the green trace $I(D_2)$conducts strictly during the narrow valley when the red trace V(n002)is LOW (0 V).
   - At the falling edge of V(n002), Pin 7 (`DIS`) shorts to Ground and discharges $C_1$from 8.0 Vthrough the smaller potentiometer segment $R_B = 11.75\ \text{k}\Omega$, producing a sharp current spike:
   $$I(D_2)_{peak} = \frac{8.0\text{ V} - 0.6\text{ V}}{11.75\text{ k}\Omega} = \frac{7.4\text{ V}}{11.75\text{ k}\Omega} \approx 629.8\ \mu\text{A}$$
   - Over the brief 0.81 msLOW interval, $I(D_2)$decays rapidly down to $\approx 289.4\ \mu$Auntil V_{C1}reaches 4.0 V, at which point V(n002)snaps back HIGH and the cycle repeats.

CHAPTER 5: HARDWARE IMPLEMENTATION



5.1 BREADBOARD ASSEMBLY PROCEDURE

Following simulation verification, the physical prototype was assembled on a solderless breadboard powered by a 12 VDC adapter (where the positive wire connects to the $+12$ Vrail and the negative wire serves as the 0 VGround reference):

1. Power & IC Connections: The `NE555` (U_1) was placed across the central breadboard channel. Pin 8 (V_{CC}) and Pin 4 (`RST`) were tied to the $+12$ Vrail, and Pin 1 (`GND`) was wired directly to the 0 VGround rail. Bypass capacitor $C_3$was placed from Pin 5 (`CV`) to Ground, and bulk capacitor $C_2$was placed across the power rails.
2. Diode-Steered Timing Network: Resistor R_2(1 kΩ) was connected from $+12$ Vto Pin 7 (`DIS`). Steering diodes $D_1$and D_2were wired in opposite polarities between Pin 7 and the outer pins of the 47 kΩpotentiometer (U_2). The middle wiper pin of $U_2$was connected to Pin 2 (`TRIG`), Pin 6 (`THRS`), and the top leg of timing capacitor C_1(100 nF to Ground).
3. MOSFET Gate Drive & Load Stage: Pin 3 (`OUT`) was connected through gate resistor R_1($100\ \Omega$) to the Gate of the `IRFZ44N` MOSFET (M_1), with pull-down resistor R_3(10 kΩ) connected from Gate to Ground and the Source pin tied to Ground. The 12 VDC motor,

reverse-biased flyback diode D_3(`1N4007`), and LED + $470\ \Omega$resistor (R_4) were connected in parallel between the $+12$ Vrail and the MOSFET Drain.

5.2 HARDWARE DEBUGGING AND MEASUREMENT METHODOLOGY

- Continuity Verification: Before applying power, a digital multimeter (DMM) in continuity mode was used to verify isolation between $+12$ Vand Ground and confirm the Pin 2–Pin 6–Wiper junction.
- Eliminating Floating-Gate Triggering: Adding the $10\ \text{k}\Omega$gate pull-down resistor (R_3) ensured stable turn-off without erratic startup conduction.
- Measuring Current and Voltage: Average motor voltage was measured with the DMM in DC Volts mode across the motor terminals, while branch currents were measured by breaking the circuit loop and inserting the DMM in DC Amperes mode in series.

CHAPTER 6: RESULTS AND DISCUSSIONS

6.1 QUANTITATIVE PERFORMANCE ACROSS DUTY CYCLE SETTINGS

Table 6.1 summarizes the theoretical, simulated, and experimental performance across five potentiometer wiper settings ($V_{CC} = 12.0$ V, $R_{tot} = 47$ kΩ, $R_2 = 1$ kΩ, $C_1 = 100$ nF).

Table 6.1: Comparison of Theoretical, Simulated, and Hardware Results

Wiper (w)	R_A (kΩ)	R_B (kΩ)	T_{ON} (ms)	T_{OFF} (ms)	Freq. (Hz)	Duty Cycle D	V_{avg} (V)	Motor & LED State
0.90	42.30	4.70	3.00	0.33	300.3	90.2%	10.82 V	Bright LED; Near-Max RPM
0.75	35.25	11.75	2.51	0.81	300.6	75.5%	9.06 V	High Brightness; Fast RPM
0.50	23.50	23.50	1.70	1.63	300.6	51.0%	6.12 V	Medium LED; Half Speed
0.25	11.75	35.25	0.88	2.44	300.6	26.6%	3.19 V	Dim LED; Slow Crawl
0.10	4.70	42.30	0.40	2.93	300.3	11.9%	1.43 V	Very Dim LED; Threshold RPM

6.2 KEY OBSERVATIONS AND ENGINEERING DISCUSSION

1. Charge Conservation Across D_1 and D_2: At $w = 0.75$, the charging duration ($T_{ON} = 2.51$ ms) is roughly three times longer than the discharging duration ($T_{OFF} = 0.81$ ms). Because the exact same charge $\Delta Q = C_1 \cdot \Delta V = 100$ nF $\times$ (8 V $-$ 4 V) $= 0.4\ \mu$Cmust enter and exit capacitor C_1 every cycle, the shorter discharge interval produces a peak current three times higher ($I(D_2)_{peak} \approx 600\ \mu$A vs. $I(D_1)_{peak} \approx 200\ \mu$A), as directly verified by the green and blue traces in the LTspice plot.
2. High Thermal Efficiency at 12 V: Operating from a 12 Vadapter drives the MOSFET gate with V(n002) ≈ 10.5 V–12 V, fully saturating M_1 so that minimal heat is dissipated even during continuous operation.

CHAPTER 7: CONCLUSION

In this micro project, an NE555-based Pulse Width Modulation (PWM) DC Motor Speed Controller was successfully designed, mathematically analyzed, simulated in LTspice, and implemented on a physical breadboard. By integrating a diode-steered potentiometer timing network (D_1, D_2, U_2) around the astable NE555 core (U_1), independent control over the charging and discharging intervals of a $100\ \mathrm{nF}$timing capacitor (C_1) was achieved, resulting in smooth duty cycle variation from $2.1\%$to 97.9%at a constant frequency of $\approx 300\ \mathrm{Hz}$.

The LTspice transient simulation waveforms confirmed the exact theoretical relationship between the rectangular PWM output voltage $V(\mathrm{n002})$and the alternating steering diode currents $I(D_1)$and $I(D_2)$. Practical hardware protections—including Pin 5 noise decoupling (C_3), MOSFET gate current limiting (R_1), gate pull-down stabilization (R_3), and inductive back-EMF clamping (D_3)—ensured reliable, spike-free hardware operation on a $12\ \mathrm{V}$DC supply.

REFERENCES

[1] H. R. Camenzind, Designing Analog Chips, Virtualbookworm Publishing (originally Signetics NE555 Application Notes, 1972), ch. 11, pp. 11-1–11-8, 2005.

[2] R. L. Boylestad and L. Nashelsky, Electronic Devices and Circuit Theory, 11th ed. Upper Saddle River, NJ, USA: Pearson Education, 2012, pp. 765–778.

[3] A. S. Sedra and K. C. Smith, Microelectronic Circuits, 7th ed. New York, NY, USA: Oxford University Press, 2014, pp. 1140–1152.

[4] M. H. Rashid, Power Electronics: Devices, Circuits, and Applications, 4th ed. Harlow, UK: Pearson, 2014, pp. 312–328.

[5] B. K. Bose, Power Electronics and Motor Drives: Advances and Trends, 2nd ed. San Diego, CA, USA: Academic Press / IEEE Press, 2020, pp. 245–260.